Dr. Dipankar Bhattacharya, Ph.D.

Marie Skłodowska-Curie Postdoctoral Fellow | Robotics for Healthcare and Manufacturing
Dyson School of Design Engineering, Imperial College London, United Kingdom

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Research Profile

Independent robotics researcher developing model-based and learning-enabled robotic systems for healthcare, rehabilitation, and advanced manufacturing. My work integrates soft robotics, cable-driven actuation, deformable-object manipulation, perception, and control to deliver experimentally validated systems with translational potential. As a Marie Skłodowska-Curie Postdoctoral Fellow at Imperial College London, I am leading the CASREx project on cable-driven soft reconfigurable exoskeletons for muscle-impairment rehabilitation, building on prior work in robotic soft esophagus systems, cable-driven robots, fabric manipulation, and learning-based control.

Fellowship Highlights

Funding	Recipient, **Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship** (EUR 276,187.92) for the project ***CASREx***, hosted at Imperial College London.
Research	Developing an independent research programme at the intersection of soft robotics, cable-driven systems, human-centred rehabilitation robotics, deformable-object manipulation, and learning-enabled control.
Leadership	Supervising MEng placement students and mentoring undergraduate, postgraduate, and PhD researchers across Imperial College London, CUHK, TransGP, and the University of Auckland.
Visibility	Invited speaker at Imperial College London, World Robotics Conference 2025, and IEEE ICMA 2025; active reviewer for RA-L, ICRA, and the ASME Journal of Mechanisms and Robotics.
Translation	Research spans healthcare rehabilitation, assistive robotics, garment manufacturing, and inspection, combining modelling, control, perception, and real-world validation.

Grants and Scholarships

Grants/Fellowships

2025	Recipient, ***Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship*** for the project "CASREx" (Grant No. 101205678), hosted at ***Imperial College London***. Funded under HORIZON-MSCA-2024-PF-01-01. Duration: 24 months; value: ***EUR 276,187.92***.
2023	Co-authored Innovation and Technology Fund (ITF), Partnership Research Programme (PRP) grant.
2022	Co-authored Research Talent Hub for ITF project (RTH-ITF) grant.
2018	Recipient, Dynamic Testing of Implant Stents Using Robotic Soft Esophagus (RoSE), BvW Holding, Cham. Duration: 12 months; value: ***NZD 39,000***.
2018, 2019	Recipient, Riddet Travel Grant, Riddet Institute Centre of Research Excellence (NZD 3,000).

Scholarships

2019–2020 Recipient, Doctor of Philosophy Scholarship, University of Auckland.

2016–2019 Recipient, Doctor of Philosophy Scholarship, Riddet Institute Centre of Research Excellence.

2011–2013 Recipient, Master of Technology Scholarship, Ministry of Human Resource Development (MHRD), Government of India.

Employment History

2025 – Present **Marie Skłodowska-Curie Postdoctoral Fellow**, Dyson School of Design Engineering, Imperial College London, United Kingdom. Supervisor: Prof. Thrishantha Nanayakkara (Morph Lab), 01/11/2025 – Present.
CASREx: Cable-driven Arm Soft Reconfigurable Exoskeleton for Muscle-Impairment Rehabilitation.

- Leading an independent fellowship project on human-inspired stiffness modulation for cable-driven elbow exosuits for Parkinson rehabilitation.
- Integrating human experiments, EMG data, digital twins, and reinforcement learning to design adaptive rehabilitation robotic systems.
- Developing and validating a cable-driven manipulator-exosuit platform with compliant actuation and antagonistic co-contraction control.

2026 – Present **Assistant Supervisor**, Dyson School of Design Engineering, Imperial College London, United Kingdom, 15/01/2026 – Present.

- Supervising two Year 3 MEng Design Engineering placement students on exosuit development projects.
- Delivered invited lecture on **Robotics Simulation and Control using PyDrake** for **DESE 71002 Robotics Research Projects**.

2024 – 2025 **Senior Research Engineer–Control Systems**, Center for Transformative Garment Production (TransGP). Supervisor: Prof. Kazuhiro Kosuge, 01/05/2025 – 31/10/2025.
Physics-based cloth manipulation with policies generated from imitation learning.

- Developed learning-based perception and imitation-learning-based control for robotic fabric manipulation.
- Supervised development and validation of robotic systems for fabric sewing and manipulation.
- Translated industrial requirements into research deliverables and contributed to research articles.
- Mentored and supervised 3 PhD students; reviewed and edited manuscripts.

2024 – 2025 **Visiting Research Associate**, Department of Electrical and Electronic Engineering, The University of Hong Kong, 02/07/2024 – 31/10/2025.

2021 – 2024 **Postdoctoral Fellow**, Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong. Supervisor: Prof. Darwin Lau, 01/03/2021 – 30/06/2024.
Project 1: Modelling and control of cable-driven robots with cable wrapping around obstacles and links.
Project 2: Design of a cable-driven growing vine robot for safer false-ceiling inspections.

- Conducted independent and collaborative research on cable-driven and vine robotics.
- Wrote research articles, proposals, and technical reports.
- Mentored undergraduate and postgraduate students.

2021 – 2022 **Visiting Postdoctoral Fellow**, Laboratoire des Sciences du Numérique de Nantes (LS2N), École Centrale de Nantes. Supervisors: Prof. Stephane Caro and Prof. Darwin Lau, 26/10/2021 – 05/01/2022.
Project: Iterative-learning control of cable-driven parallel robots.

2017 – 2020 **Graduate Teaching Assistant**, The University of Auckland, New Zealand, 2017–2020.

2018 – 2019 **Research Assistant** at The University of Auckland, under the direction of Prof Peter Xu, 09/01/2018 – 08/01/2019.

2013 – 2016 **Lecturer (Assistant Professor Grade-1)**, School of Electrical, Electronics and Communication Engineering, Galgotias University, India, 19/07/2013–18/06/2016.

Research Publications

Journal (Published/Accepted)

[1] B. Jin*, A. Kobayashi*, D. Bhattacharya, A. Seino, F. Tokuda, N. Tien, and K. Kosuge. "Automated Straight-line Sewing of Stretchable Fabrics with Different Lengths". In: *IEEE Robotics and Automation Letters* (2025). *Equal contribution., pp. 1–8. DOI: 10.1109/LRA.2025.3619791.

[2] D. Bhattacharya, T. K. Cheung, Y. Wang, and D. Lau. "Kinematic and Dynamic Modeling of Cable-Object Interference and Wrapping in Complex Geometrical-Shaped Cable-Driven Parallel Robots". In: *Mechanism and Machine Theory* 214 (2025), p. 106092. DOI: 10.1016/j.mechmachtheory.2025.106092.

[3] D. Bhattacharya, Y. P. Chan, S. Shang, Y. S. Chan, Y. Tan, and D. Lau. "Tri-Space Operational Control of Redundant Multilink and Hybrid Cable-Driven Parallel Robots Using an Iterative-Learning based Reactive Approach". In: *IEEE Transactions on Control Systems and Technology* (2023). DOI: 10.1109/TCST.2023.3263689.

[4] D. Bhattacharya, L. K. Cheng, and W. Xu. "Nonlinear Model Predictive Control of a Robotic Soft Esophagus for Endoprosthetic Stent Testing." In: *IEEE Transactions on Industrial Electronics* (2021). DOI: 10.1109/TIE.2021.3121755.

[5] D. Bhattacharya, S. Jesna, L. K. Cheng, and W. Xu. "RoSE: A Robotic Soft Esophagus for Endoprosthetic Stent Testing." In: *Soft Robotics* (2021). DOI: 10.1089/soro.2019.0205.

[6] Y. Dang, Y. Liu, R. Hashem, D. Bhattacharya, J. Allen, M. Stommel, L. K. Cheng, and W. Xu. "SoGut: A Soft Robotic Gastric Simulator". In: *Soft Robotics* (2021). DOI: 10.1089/soro.2019.0136.

[7] D. Bhattacharya, L. K. Cheng, and W. Xu. "Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot". In: *IEEE Transactions on Industrial Electronics* 67.6 (2020), pp. 4711–4720. DOI: 10.1109/TIE.2019.2928239.

[8] D. Bhattacharya, M. G. Nisha, and G. Pillai. "Relevance vector-machine-based solar cell model". In: *International Journal of Sustainable Energy* 34.10 (2015), pp. 685–692. DOI: 10.1080/14786451.2014.885030.

Books and Chapters

[9] D. Bhattacharya, L. K. Cheng, S. Dirven, and W. Xu. *Artificial Intelligence Approach to the Trajectory Generation and Dynamics of a Soft Robotic Swallowing Simulator*. Cham: Springer International Publishing, 2019, pp. 3–16. DOI: 10.1007/978-3-319-78452-6_1.

Conference Proceedings

[10] W. Dong, D. Bhattacharya, A. Kobayashi, A. Seino, F. Tokuda, X. Huang, K. Tang, N. C. Tien, and K. Kosuge. "Precise Top-layer Fabric Segmentation for Fabric Destacking with Edge- and Shape-Aware Deep Networks". In: *2025 IEEE International Conference on Mechatronics and Automation (ICMA)*. 2025, pp. 1343–1348.

[11] D. Bhattacharya, L. K. Cheng, and W. Xu. “Testing and Analysis of Migration Displacement of a Flared Stent Deployed in an Esophageal Swallowing Robot.” In: *25th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*. 2018, pp. 1–6. DOI: 10.1109/M2VIP.2018.8600911.

[12] D. Bhattacharya, L. K. Cheng, S. Dirven, and W. Xu. “Actuation planning and modeling of a soft swallowing robot”. In: *24th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*. 2017, pp. 1–6. DOI: 10.1109/M2VIP.2017.8211476.

In Review/Preparation

[13] D. Bhattacharya*, K. Tang*, X. Hang, F. Tokuda, N. Tien, and K. Kosuge. *RTFF: Random-to-Target Fabric Flattening Policy Using Dual-Arm Manipulator, IEEE International Conference on Robotics and Automation (ICRA)*. Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*. in review. *Equal contribution. 2026.

[14] W. Wang, D. Bhattacharya, A. Kobayashi, A. Seino, F. Tokuda, N. Tien, and K. Kosuge. *Automated Fabric Alignment Using Global-Local Optimization and Point Cloud Registration, IEEE Transactions on Automation Science and Engineering (T-ASE)*. in review.

[15] T. Cheung, D. Bhattacharya, and D. Lau. *Rapidly Deployable Deformable Frame Cable-Driven Parallel Robots with Small Footprint*. in preparation.

[16] Y. Wang, D. Bhattacharya, and D. Lau. *Static Modelling and Workspace Analysis with Arbitrary Cable Attachment Design in a Cable-Driven Parallel Robot with Passive Backbone, IEEE Transactions on Automation Science and Engineering*. in preparation.

Academic Awards and Professional Service

Academic Awards

2025 Selected, **European Talent Academy 2026** (Imperial College London, TU Munich, Politecnico di Milano).

2017 Recipient, **Best Conference Paper Award**, 5th International Conference on Robotic Intelligence Technology and Applications (RiTA), Daejeon, Korea.

2017 **Ranked 2nd** in poster presentation, Delivery of Functionality in Complex Food Systems (DoF 2017) Conference.

Competitive Examinations

2011 **Percentile: 99.12% (Rank 1,211/137,853)** in Graduate Aptitude Test in Engineering (GATE).

2010 **Percentile: 96.00%** in Graduate Aptitude Test of Engineering (GATE).

2004 **Ranked 31st** in All India Junior Mathematics Olympiad.

Professional Activities

05/02/26 Invited Speaker, **Dyson Engineering Departmental Seminar**, Imperial College London.

29/01/26 Invited Speaker, **Imperial Robotics Postgrad Committee**, Imperial College London. Talk: *Learning-based Fabric Manipulation with Mesh Perception and Visual Servoing*.

28/01/26 Delivered invited lecture on **Robotics Simulation and Control using PyDrake** for DESE71002 Robotics Research Projects, Imperial College London.

08/08/25 Invited Speaker, **World Robotics Conference (WRC 2025)**, Beijing, China. Talk: *Fabric Modelling and Perception for Garment Manufacturing*.

03/08/25 Workshop Organizer and Presenter, **IEEE ICMA 2025**, Beijing, China. Talk: *Mesh-based Fabric State Estimation and Manipulation*.

29/05/23	Poster Presentation, **IEEE ICRA 2023**, ExCeL London, United Kingdom.
2018	Presenter, **BioMouth 2018 Meet**, Auckland, New Zealand.
2021 – Present	Reviewer for *IEEE Robotics and Automation Letters* (RA-L), *IEEE International Conference on Robotics and Automation* (ICRA), and *ASME Journal of Mechanisms and Robotics*.
2017 – Present	Member, Institute of Electrical and Electronics Engineers (IEEE).
2014	Former member of the Electrical Engineering Society (EES), IIT Roorkee, and NERIST Electronics Society (NES), NERIST.
2006	Awarded an Outstanding Certificate by the National Service Scheme (NSS) for contributions to blood donation camps, village cleanliness programmes, and children's education.

Teaching Experience

2026	**DESE71002**, *Robotics Research Projects*, Year 4/MSc (Design Engineering), Dyson School of Design Engineering, Imperial College London, United Kingdom. Guest lecture on *Robotics Simulation and Control using PyDrake* (28/01/2026).
2021 – 2024	**MAEG5090**, *Topics in Robotics*, for postgraduate students (MSc/MPhil), Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Hong Kong SAR. Class size: 50 students.
2017 – 2020	Undergraduate teaching at The University of Auckland, New Zealand: **MECHENG 706** *Mechatronics Design Projects*; **MECHENG 306** *Design of Sensing and Actuating Systems*; **ENGGEN 115** *Principles of Engineering Design*; **ENGGEN 131** *Introduction to Engineering Computation and Software Development*.
2013 – 2016	Undergraduate courses (class size: 60 each), School of Electrical, Electronics and Communication Engineering, Galgotias University, India: **BECE3304** *Neural Networks and Fuzzy Control*; **BECE2010** *Digital Electronics*; **BECE2016** *Signals and Systems*; **BTEE2002** *Network Analysis and Synthesis*; **BEEE3002** *Control Systems*; **BEEE2001** *Electrical Measurement and Instrumentation*; **BECE3020** *Digital Signal Processing*; **BTEE3019** *Advanced Control System*.
2013 – 2016	Postgraduate courses (class size: 20), School of Electrical, Electronics and Communication Engineering, Galgotias University, India: **BEEE5004** *Digital Control*.

Education

2016 – 2021	**Ph.D., Mechatronics Engineering**, Department of Mechanical and Mechatronics Engineering, The University of Auckland, New Zealand, 01/08/2016 – 14/04/2021. Thesis title: *Robotic Soft Esophagus Design, Modelling, and Control for Endoprosthetic Stent Testing and Food Swallow Investigation* Developed a bio-inspired robotic soft esophagus for physiologically relevant stent testing and swallow investigation, resulting in journal articles, conference papers, and an invited book chapter.
2011 – 2013	**M.Tech., Systems and Control**, Department of Electrical Engineering, Indian Institute of Technology Roorkee, India, 22/07/2011 – 04/07/2013. Dissertation title: *Non-Linear Predictive Control Model using Relevance Vector Machine*. GPA: **8.4 / 10.0** Developed a nonlinear predictive control framework using a Relevance Vector Machine and validated it on a magnetic-levitation system, resulting in a journal article.

2004 – 2010	**B.Tech., Electronics and Communication Engineering**, North Eastern Regional Institute of Science and Technology, India, 21/07/2004 – 23/06/2010. Project title: *Design and Synthesis of Sixth Order Tow Thomas Bi-quad filter*. GPA: **9.1 / 10.0**

Skills

Languages	English and Hindi: professional reading, writing, and speaking proficiency.
Research areas	Soft robotics, cable-driven robots, deformable-object manipulation, robot modelling and control, visual servoing, learning-based control, and robotic systems for healthcare and manufacturing.
Programming	Python, MATLAB, PyTorch, ROS2, LabVIEW, Git, and LaTeX.
Simulation and tools	NVIDIA Isaac Sim, PyDrake, LeRobot, OpenSim, Onshape, Blender, SOLIDWORKS, and Autodesk Inventor.
Hardware	NVIDIA Jetson, Raspberry Pi, myRIO, and Arduino.